Rugved Deshmukh

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Education

University of Sydney — Sydney, NSW, Australia
Masters of Professional Engineering - Software Engineering (Accelarated) — *February 2026 - March 2028*

M.S. Ramiah University of Applied Sciences — Bangalore, Karnataka, India
Bachelor of Technology in Computer Science and Engineering, ***CGPA: 8.85*** — *September 2021 - July 2025*

Work Experience

Software Developer (*APAC RnD Urology*) — Oct 2023 – December 2023
Boston Scientific — *Pune, Maharashtra, India*

- Worked on Linux-based benchmarking tools and utilities for our medical devices. Created a real-time performance and resource management system for our SUDs.
- Worked with the Qt framework and UI development for devices.

UI/UX and Web Developer — Jan 2021 – October 2024
Freelance — *Pune, Maharashtra, India*

- Collaborated with clients to deliver tailored, performance-optimized websites and designs. Built responsive, cross-browser websites using modern frameworks and tools and provided ongoing website maintenance and updates.

Academic Projects

Enhancing Stress and Anxiety Detection Accuracy using Multimodal Sensor Data — August 2024 – Jan 2025
First Author, BIDA 2025, under Springer Nature, mentored by Dr. Yogesh K M — *Bangalore, Karnataka, India*

- This study aims to develop a real-time, non-invasive system for predicting stress and anxiety levels using wearable device signals (ACC, skin temperature, EDA) and machine learning, with the goal of enabling continuous mental health monitoring and personalized intervention.

Projects

Interactive Portfolio Site | *JavaScript, Three.js (WebGL), GLSL, Vite, GSAP, HTML/CSS, Vercel* — October 2026

- Designed and built an interactive 3D portfolio website in JavaScript with Three.js (WebGL), GLSL shaders, GSAP and Vite, deployed on Vercel. Optimised for performance through geometry merging, texture atlasing and glTF/Meshopt asset compression (94% smaller), and included a responsive, animated plain-HTML fallback for accessibility.

KIWIX - Youtube UI Revamp (Open Source) | *JavaScript, HTML, CSS, Python* — June 2023 – 2025

- Contributed to the frontend development using JavaScript, HTML, and CSS, while integrating Python for backend support with Kiwix as part of their GSoC projects. Enhanced cross-platform compatibility and user accessibility.

AOSSIE - PictoPy (Open Source) | *PyQT5, OpenCV, SQLite, Pytorch, PyInstaller* — July 2023 – 2025

- Aided in the development of the user interface using PyQt5. Also implemented OpenCV for image processing and PyTorch for smart tagging, categorizing photos based on objects, faces, and scenes, as part of a GSoC project under Australian Open Source Software Innovation and Education.

RPRMS | *Psutils, Bash, Python, Linux (Ubuntu)* — October 2023

- A high-efficiency real-time performance & resource monitoring system for Linux-based devices, including visualization and storage of performance data.

Voxel Engine | *OpenGL, ModernGL, Pygame, Numpy, PyGLM, Numba, OpenSimplex* — August 2023

- Developed an engine with custom shaders, clouds and fog, water transparency, world, and chunk generation.

Technical Skills

Languages: Python, C/C++, SQL, React.js, Typescript JavaScript, HTML/CSS, Bash Scripting, MATLAB
Frameworks: WordPress, Django, Unreal Engine 4, Unity, Android SDK, Qt
Developer Tools: Git, Docker, VS Code, Visual Studio, PyCharm, Eclipse
Libraries: pandas, NumPy, Matplotlib, OpenGL, ModernGL, Pygame, PyGLM
Miscellaneous: Figma, Adobe Suite, Certified Microsoft Office Specialist